

NFAs accept the Regular Languages

Equivalence of Machines

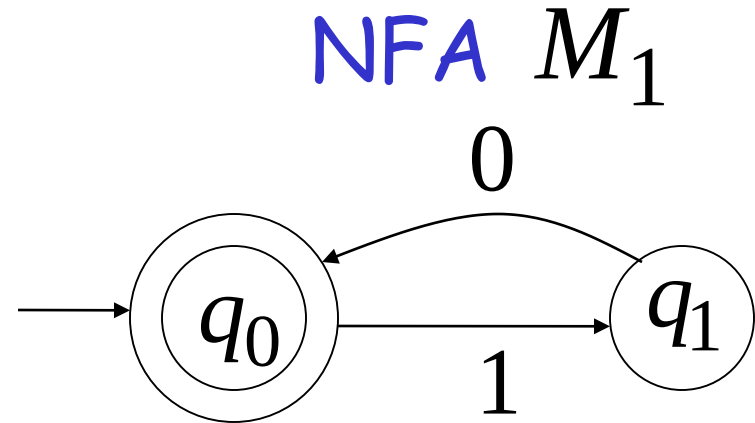
Definition for Automata:

Machine M_1 is equivalent to machine M_2

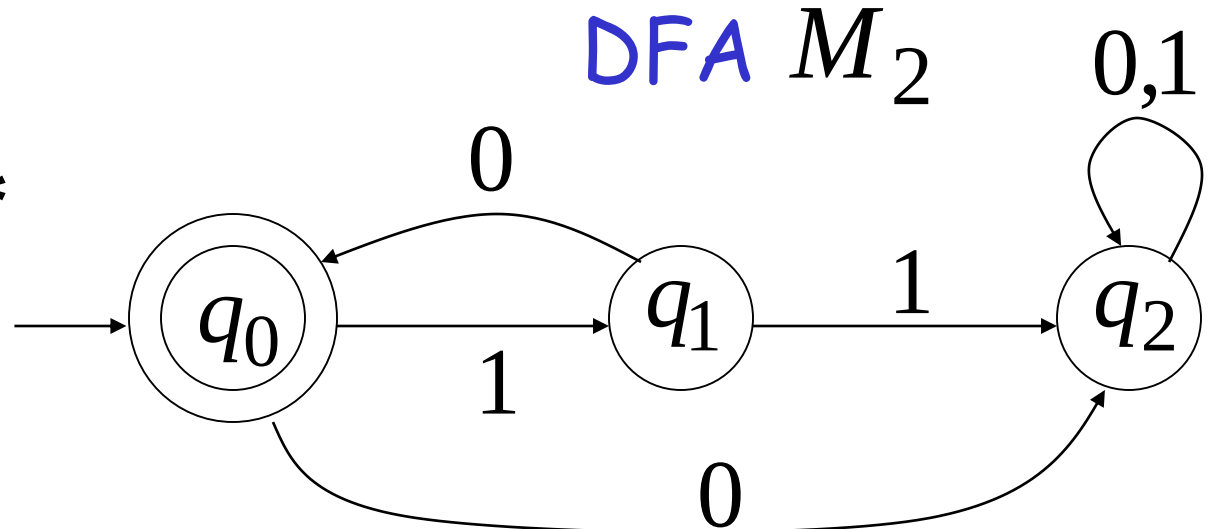
if $L(M_1) = L(M_2)$

Example of equivalent machines

$$L(M_1) = \{10\}^*$$



$$L(M_2) = \{10\}^*$$



We will prove:

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{accepted} \\ \text{by NFAs} \end{array} \right\} = \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Languages
accepted
by DFAs

NFAs and DFAs have the
same computation power

Proof: we only need to show

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{accepted} \\ \text{by NFAs} \end{array} \right\} \supseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

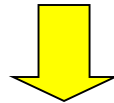
AND

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{accepted} \\ \text{by NFAs} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Step 1

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{accepted} \\ \text{by NFAs} \end{array} \right\} \supseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Proof: Every DFA is trivially an NFA

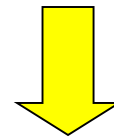


Any language L accepted by a DFA
is also accepted by an NFA

Step 2

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{accepted} \\ \text{by NFAs} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

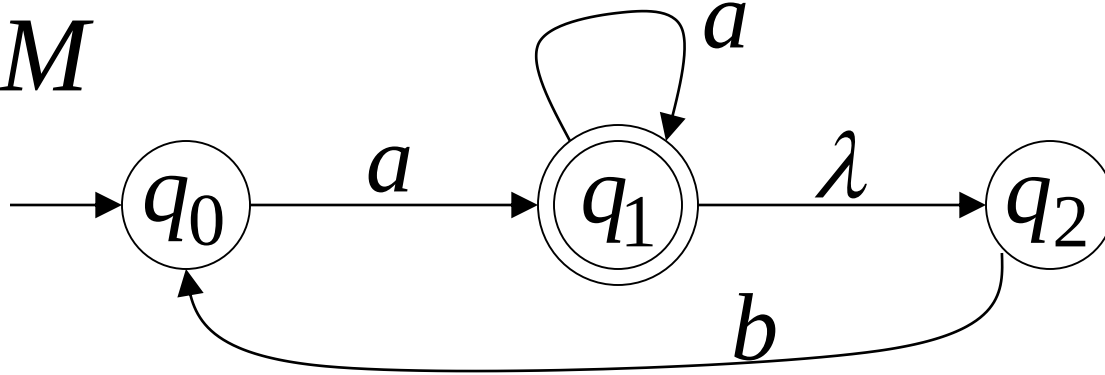
Proof: Any NFA can be converted to an equivalent DFA



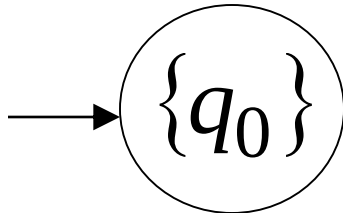
Any language L accepted by an NFA is also accepted by a DFA

Conversion NFA to DFA

NFA M

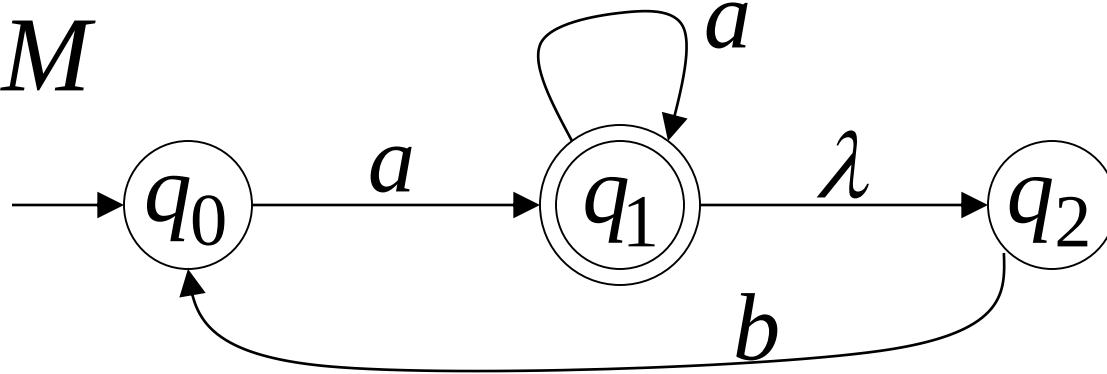


DFA M'

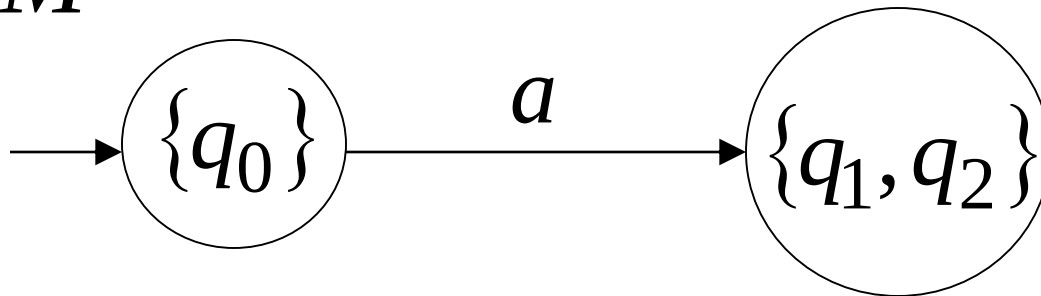


$$\delta^*(q_0, a) = \{q_1, q_2\}$$

NFA M

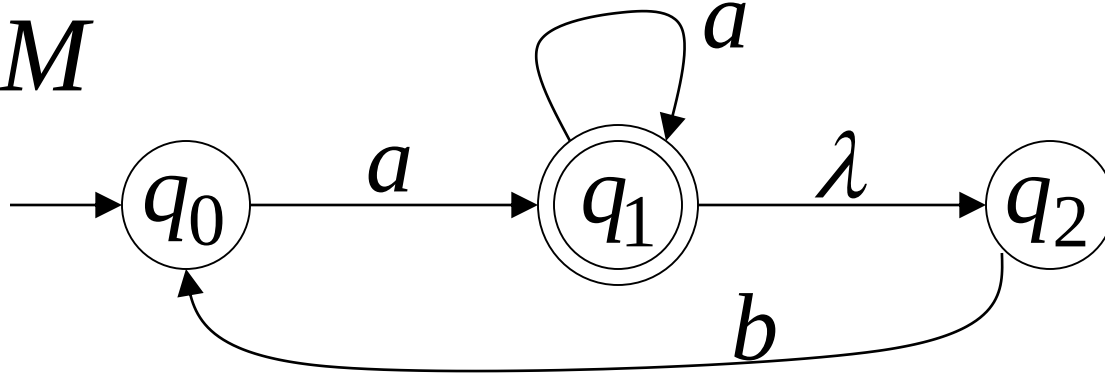


DFA M'

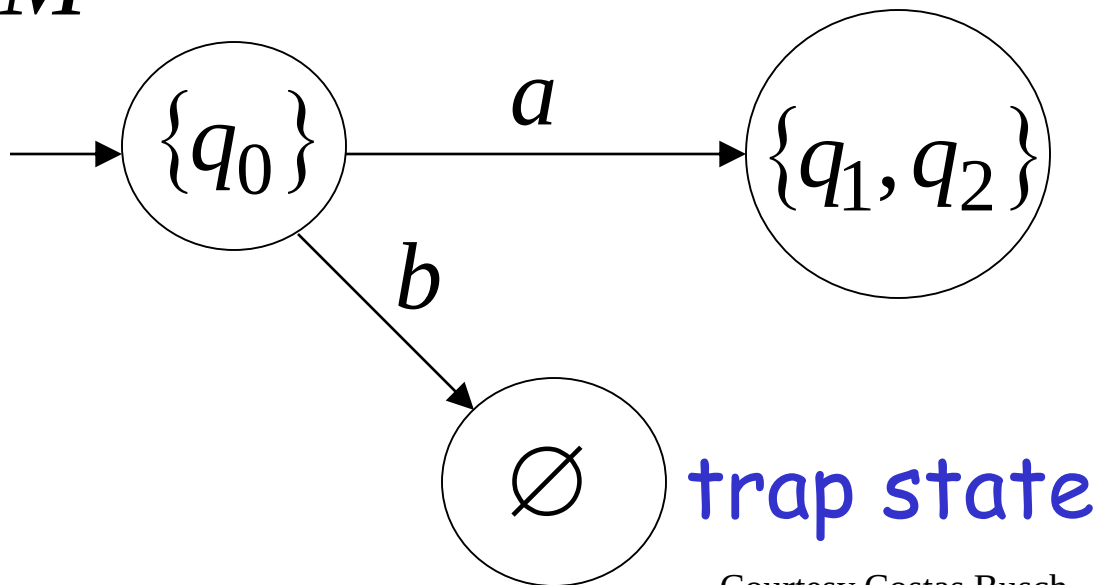


$$\delta^*(q_0, b) = \emptyset \quad \text{empty set}$$

NFA M

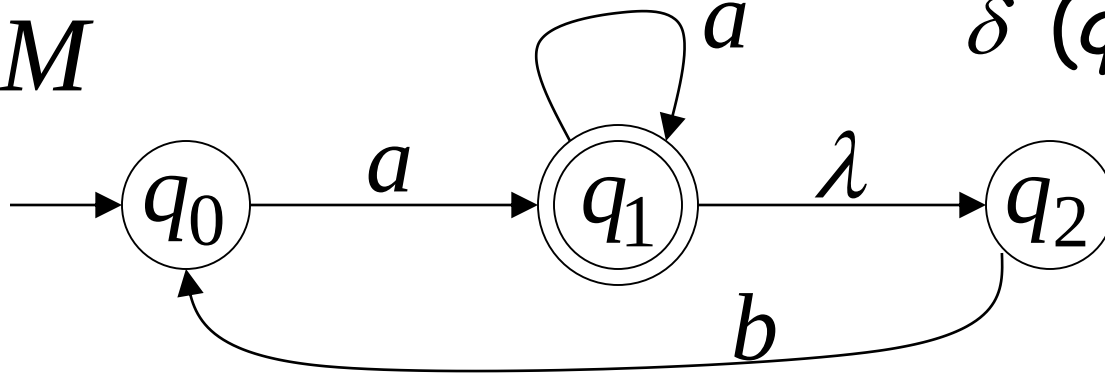


DFA M'



trap state

NFA M



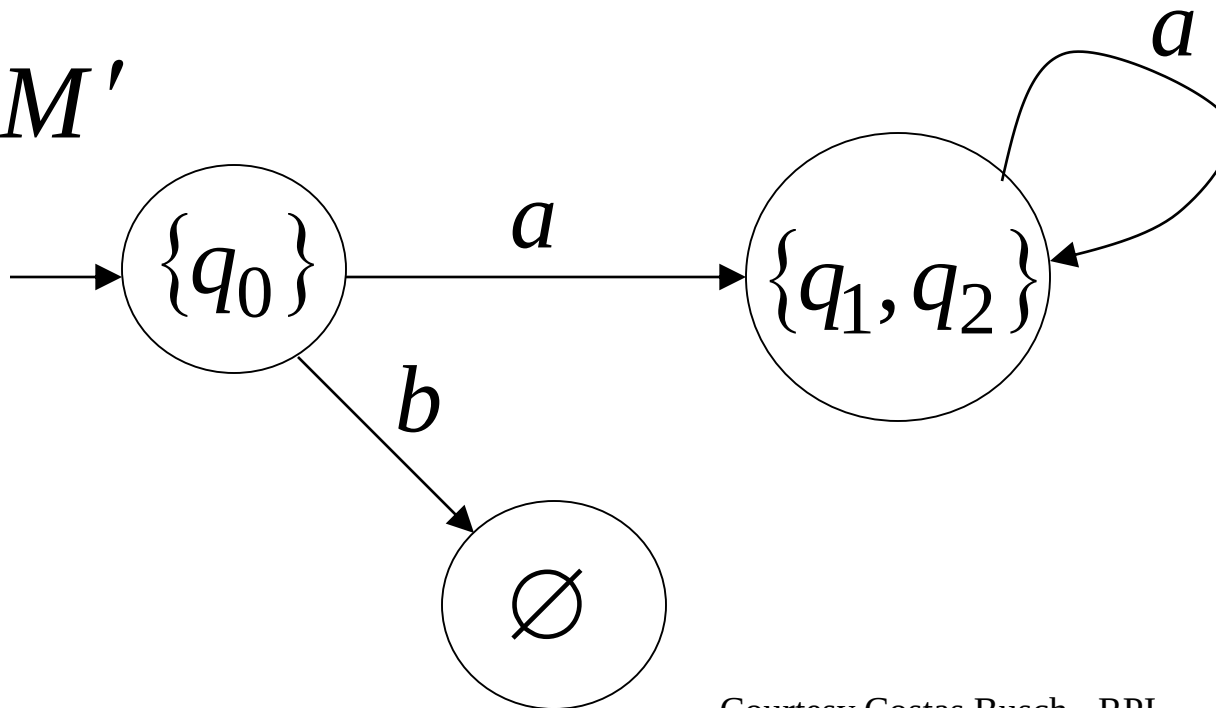
$$\delta^*(q_1, a) = \{q_1, q_2\}$$

$$\delta^*(q_2, a) = \emptyset$$

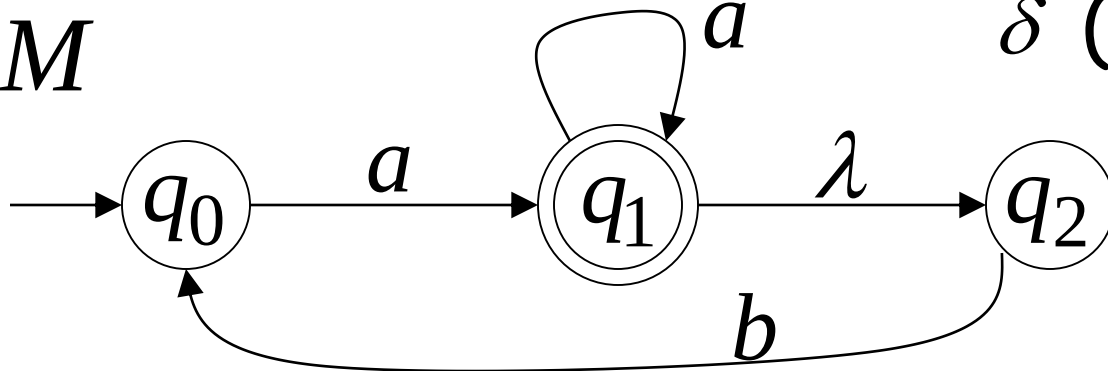
union

$$\{q_1, q_2\}$$

DFA M'



NFA M



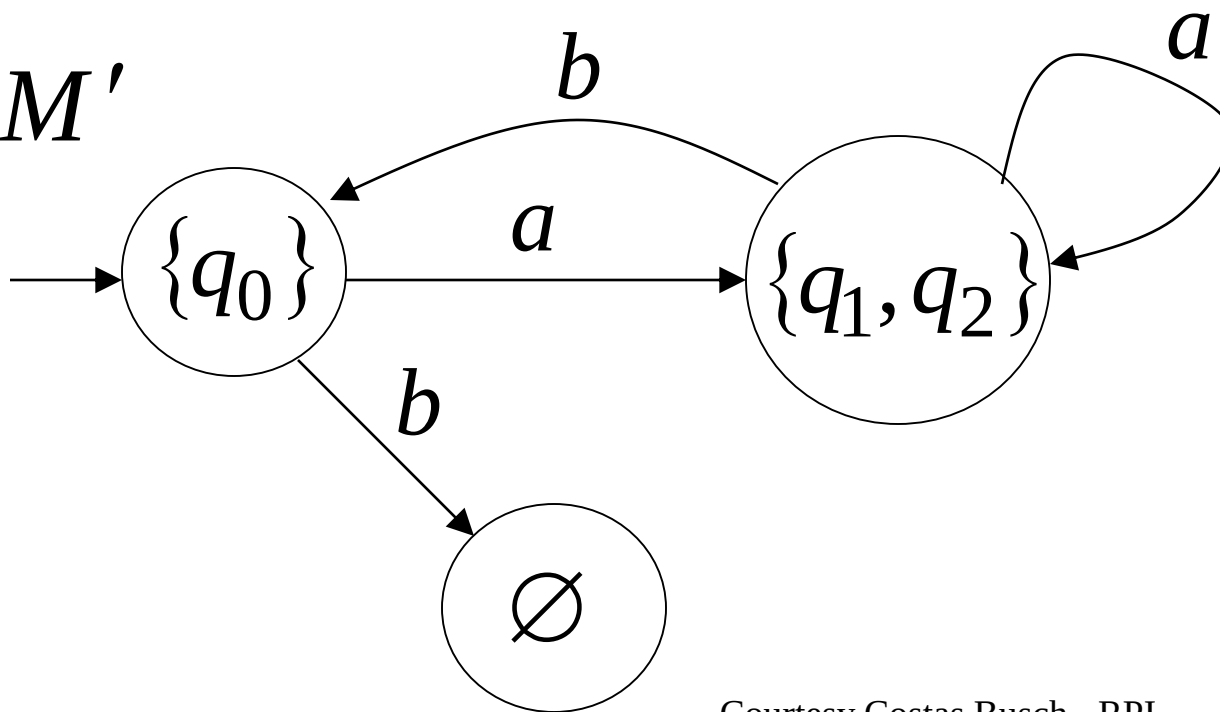
$$\delta^*(q_1, b) = \{q_0\}$$

$$\delta^*(q_2, b) = \{q_0\}$$

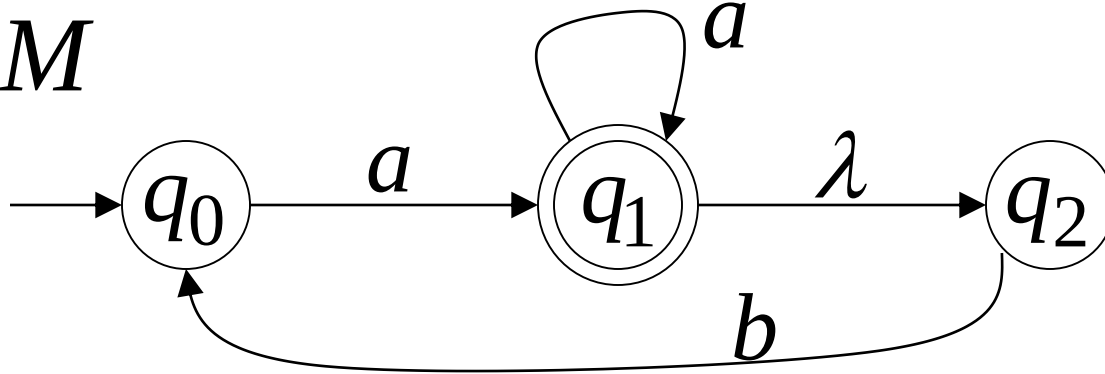
union

$$\{q_0\}$$

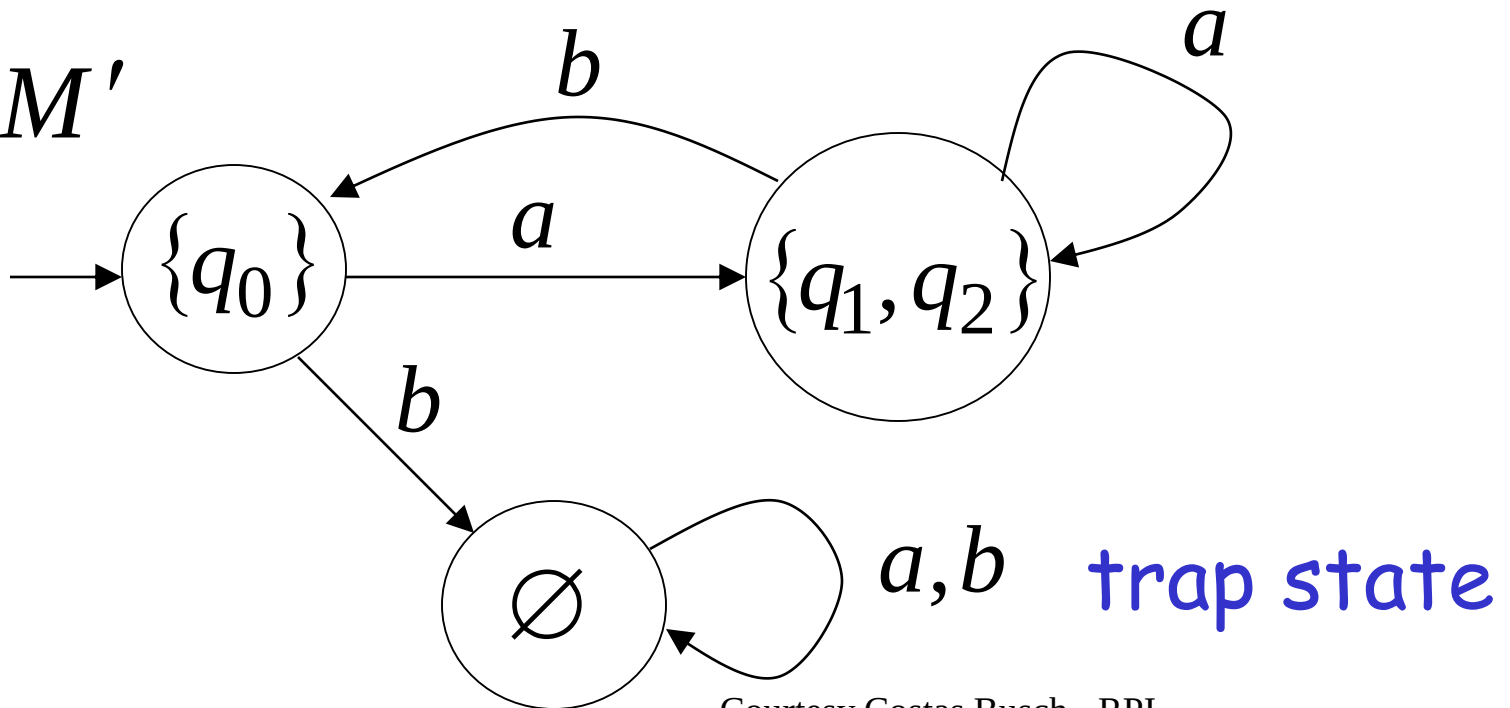
DFA M'



NFA M

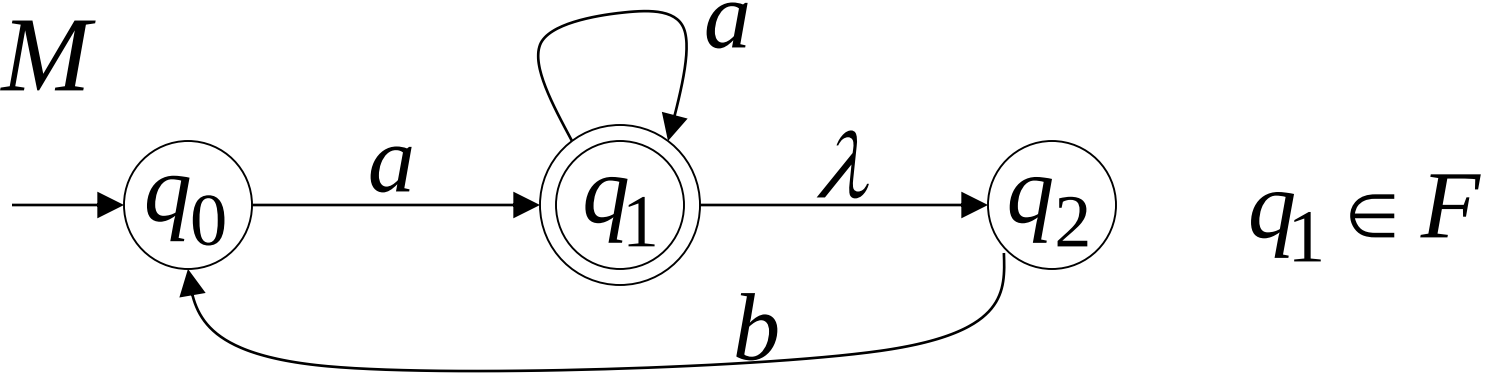


DFA M'

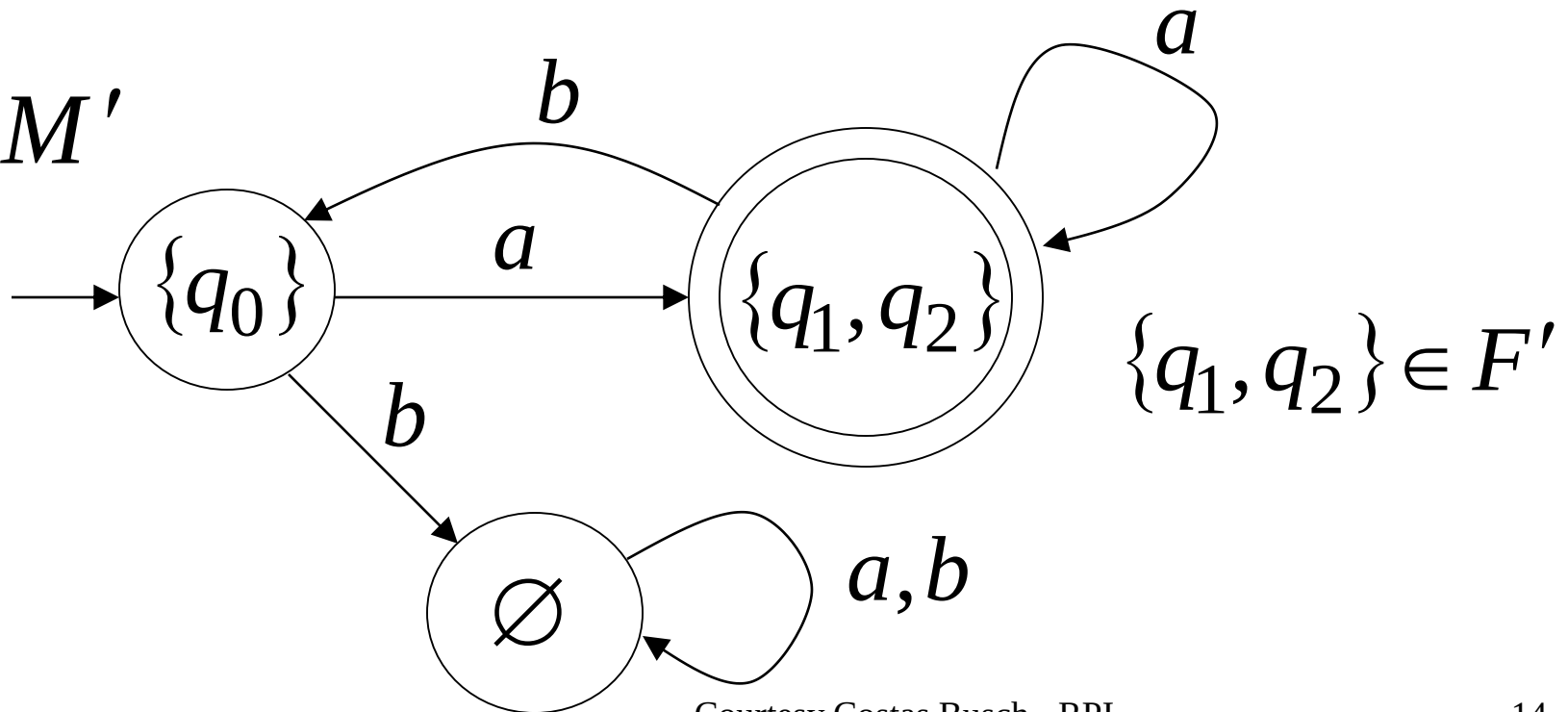


END OF CONSTRUCTION

NFA M

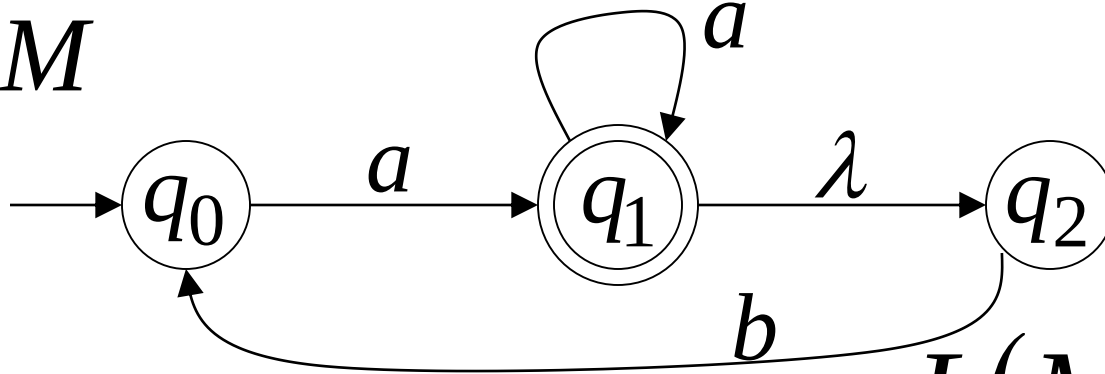


DFA M'



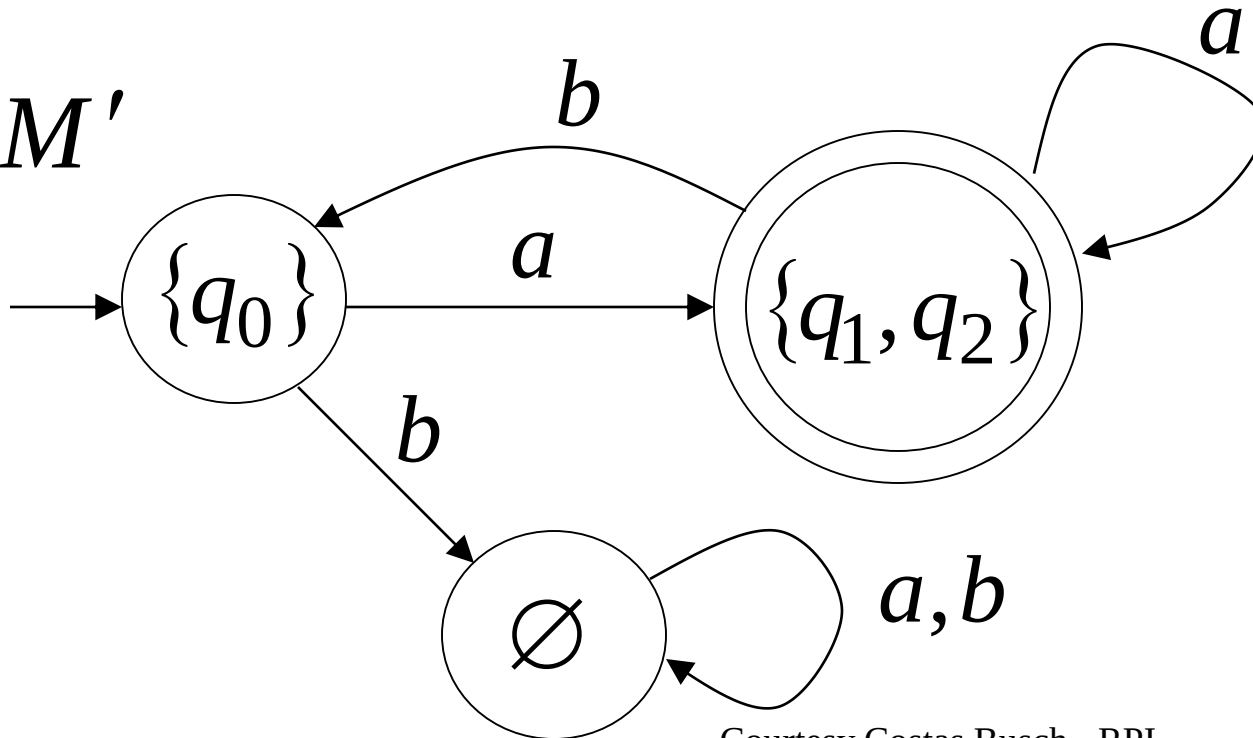
Convert NFA to DFA

NFA M



$$L(M) = L(M')$$

DFA M'



General Conversion Procedure

Input: an NFA M

Output: an equivalent DFA M'
with $L(M) = L(M')$

Procedure: subset construction

NFA to DFA: Formally

We are given an NFA $M = (Q_N, \Sigma, \delta_N, q_0, F_N)$

We want to convert it to an equivalent DFA $M' = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$

With

$$L(M) = L(M')$$

If the NFA has n states

$$Q_N = \{q_0, q_1, q_2, \dots, q_n\}$$

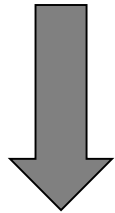
the DFA has 2^n states in the power set of
 Q_N

i.e. $\emptyset, \{q_0\}, \{q_1\}, \{q_1, q_2\}, \{q_3, q_4, q_7\}, \dots$

Procedure NFA to DFA

1. Initial state of NFA: q_0

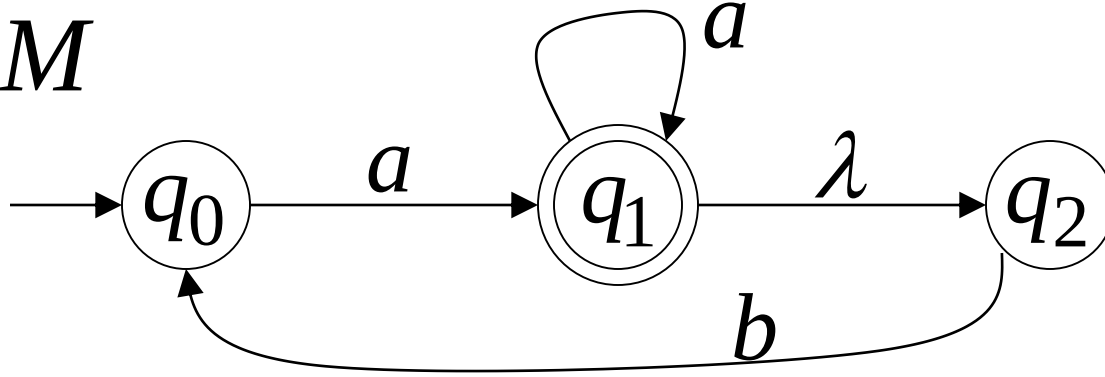
$$\delta^*(q_0, \lambda) = \{q_0, \dots\}$$



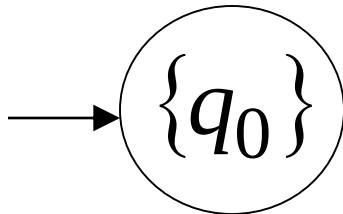
Initial state of DFA: $\{q_0, \dots\}$

Example

NFA M



DFA M'



Procedure NFA to DFA

2. For every DFA's state $\{q_i, q_j, \dots, q_m\}$

Compute in the NFA

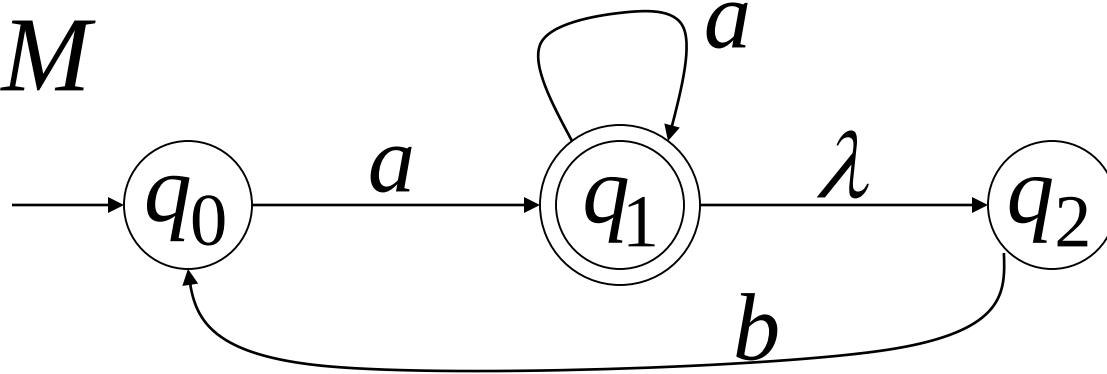
$$\left. \begin{array}{l} \delta^*(q_i, a) \\ \cup \delta^*(q_j, a) \\ \dots \\ \cup \delta^*(q_m, a) \end{array} \right\} \text{Union} = \{q'_i, q'_j, \dots, q'_m\}$$

Add transition to DFA

$$\delta(\{q_i, q_j, \dots, q_m\}, a) = \{q'_i, q'_j, \dots, q'_m\}$$

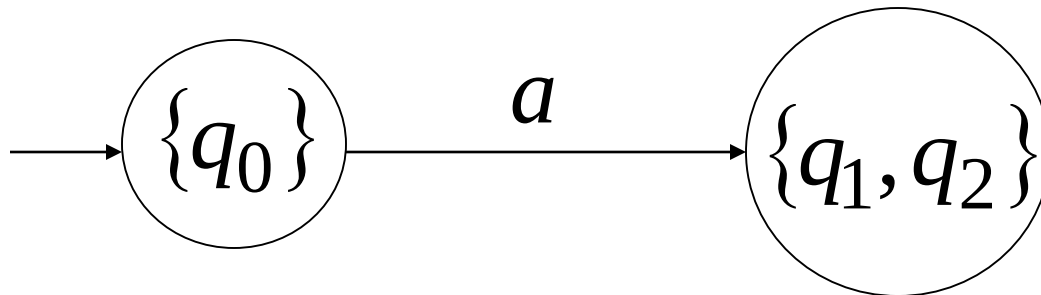
Example

NFA M



$$\delta^*(q_0, a) = \{q_1, q_2\}$$

DFA M'



$$\delta(\{q_0\}, a) = \{q_1, q_2\}$$

Procedure NFA to DFA

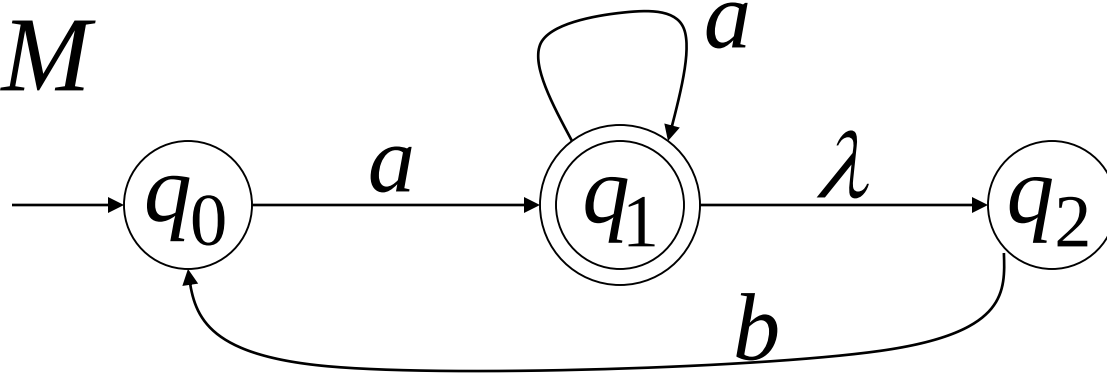
Repeat Step 2 for all letters in alphabet,
until no more transitions can be added.

OR

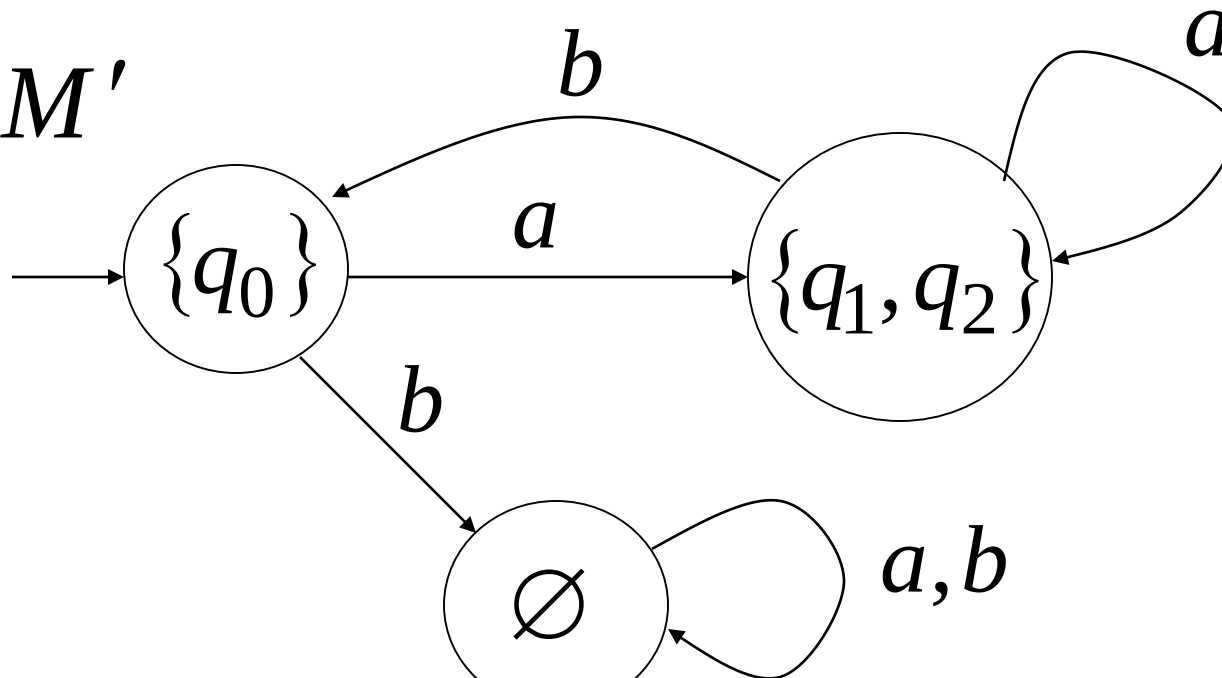
Repeat Step 2 for every state in DFA and
symbols in alphabet until no more states
can be added in the DFA

Example

NFA M



DFA M'



Procedure NFA to DFA

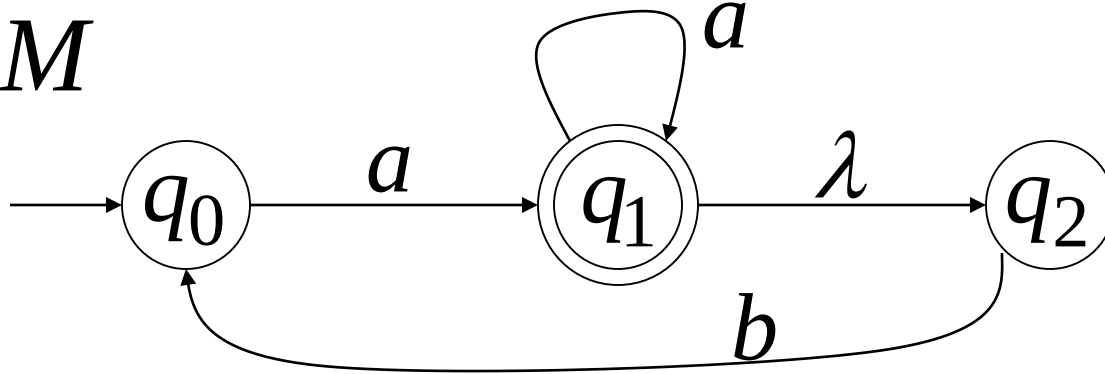
3. For any DFA state $\{q_i, q_j, \dots, q_m\}$

If some q_j is a final state in the NFA

Then, $\{q_i, q_j, \dots, q_m\}$
is a final state in the DFA

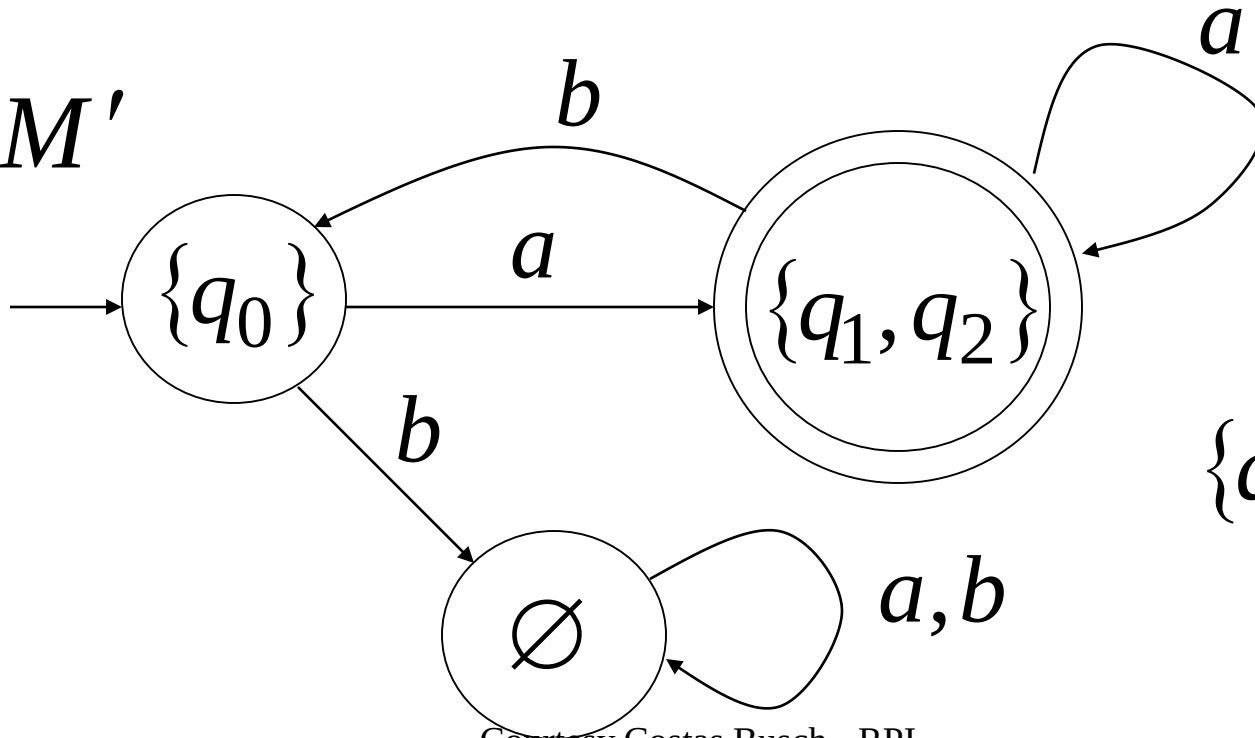
Example

NFA M



$q_1 \in F$

DFA M'



$\{q_1, q_2\} \in F'$

Lemma:

If we convert NFA M to DFA M'
then the two automata are equivalent:

$$L(M) = L(M')$$

Proof:

We only need to show: $L(M) \subseteq L(M')$

AND

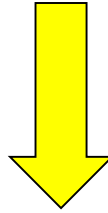
$$L(M) \supseteq L(M')$$

First we show: $L(M) \subseteq L(M')$

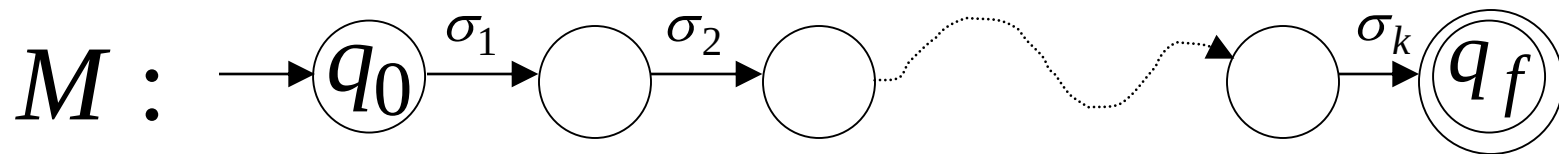
Take arbitrary: $w \in L(M)$

We will prove: $w \in L(M')$

$$w \in L(M)$$

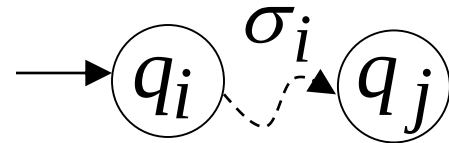


$$w = \sigma_1 \sigma_2 \cdots \sigma_k$$



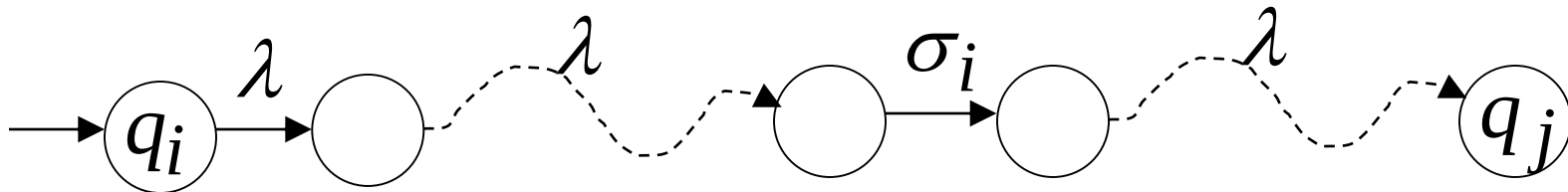
NOTE:

symbol



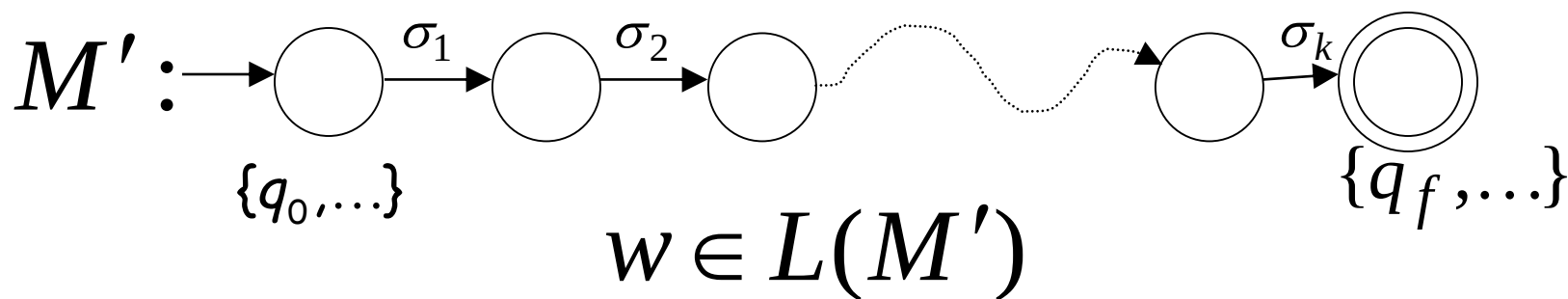
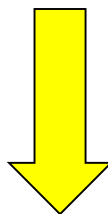
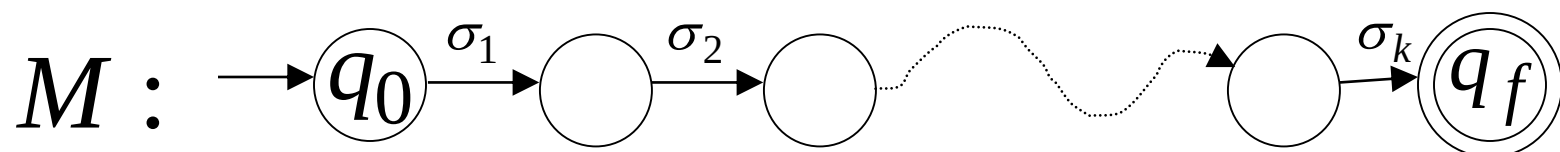
denotes a possible sub-path like

symbol



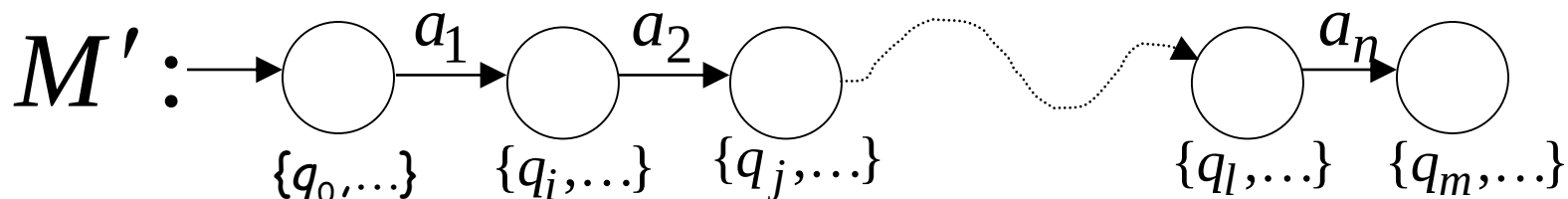
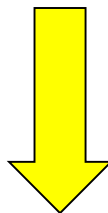
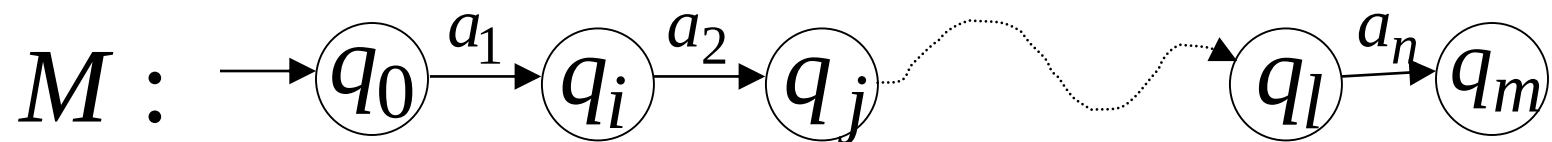
We will show that if $w \in L(M)$

$$w = \sigma_1 \sigma_2 \cdots \sigma_k$$



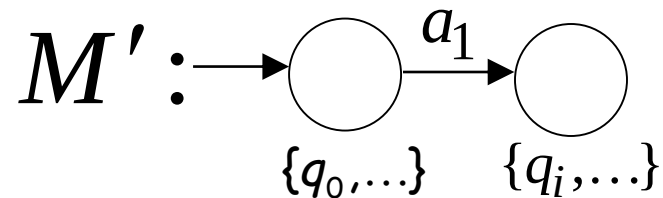
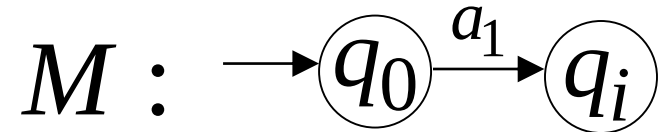
More generally, we will show that if in M :

(arbitrary string) $v = a_1 a_2 \cdots a_n$



Proof by induction on $|v|$

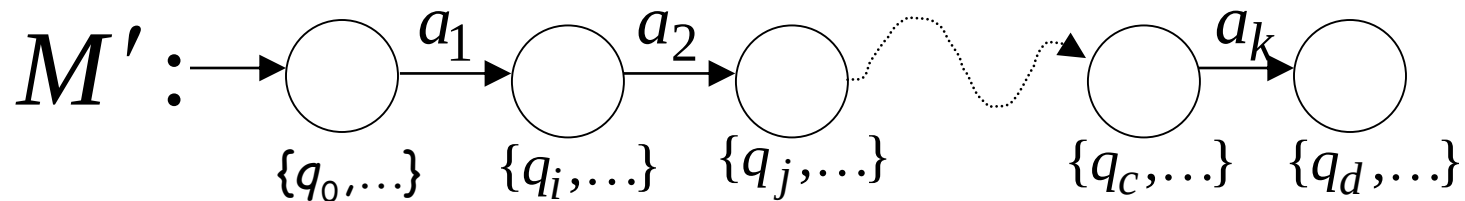
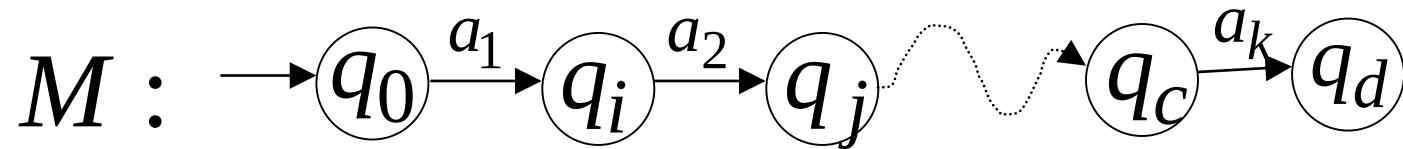
Induction Basis: $v = a_1$ $|v| = 1$



is true by construction of M'

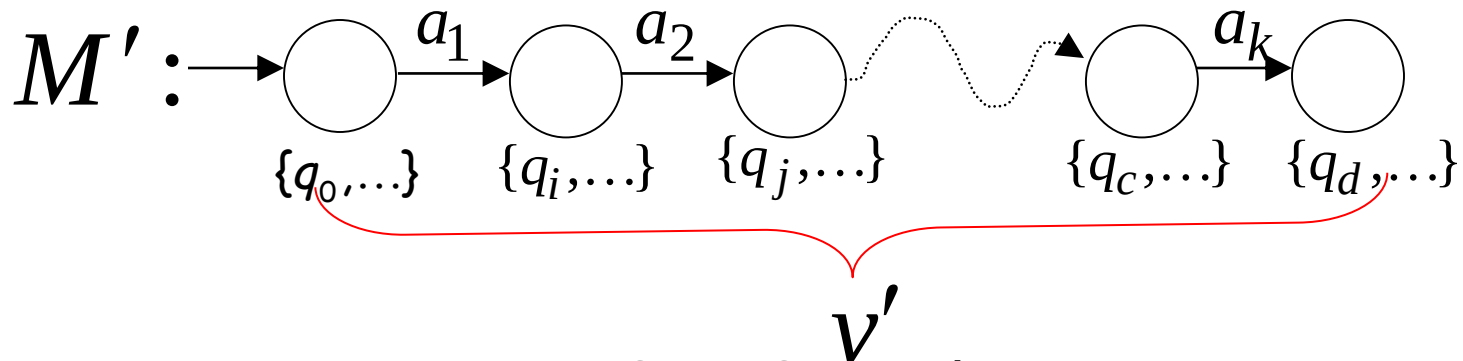
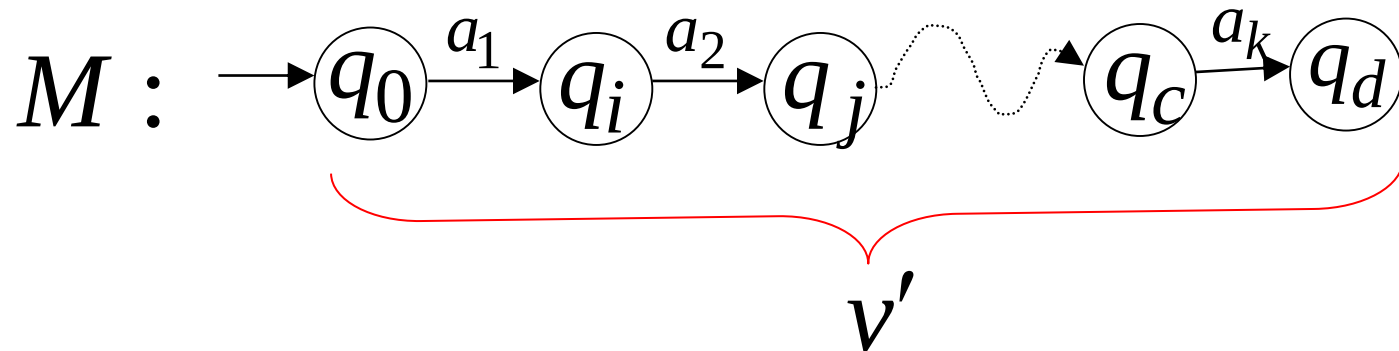
Induction hypothesis: $1 \leq |v| \leq k$

$$v = a_1 a_2 \cdots a_k$$



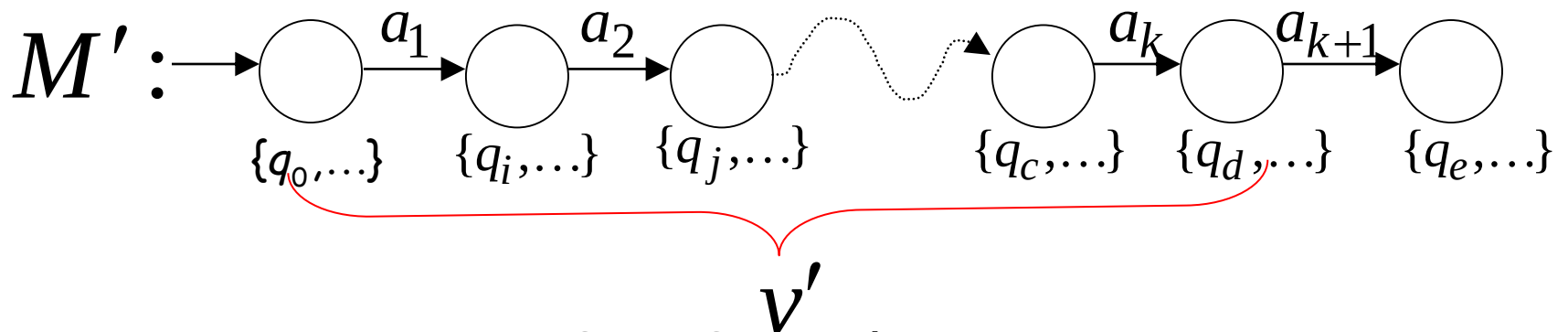
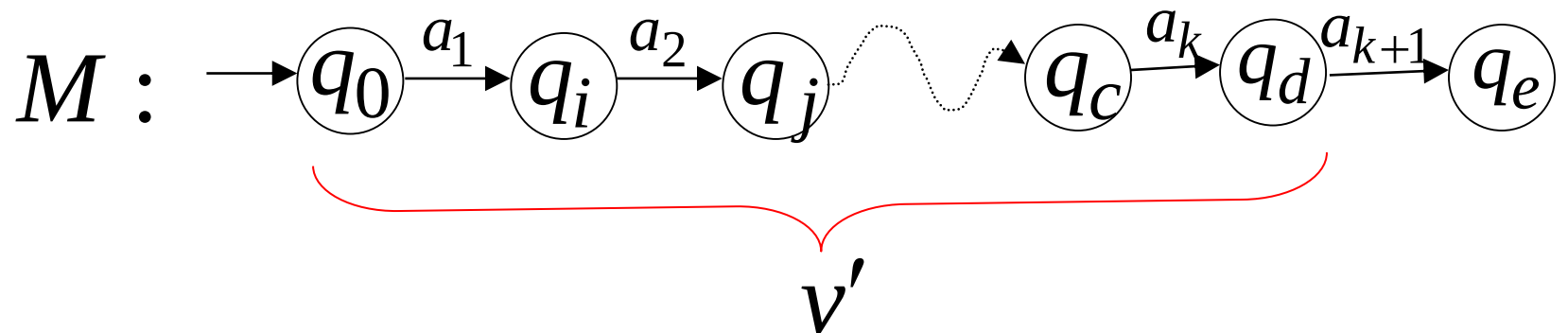
Induction Step: $|v| = k + 1$

$$v = \underbrace{a_1 a_2 \cdots a_k}_{v'} a_{k+1} = v' a_{k+1}$$



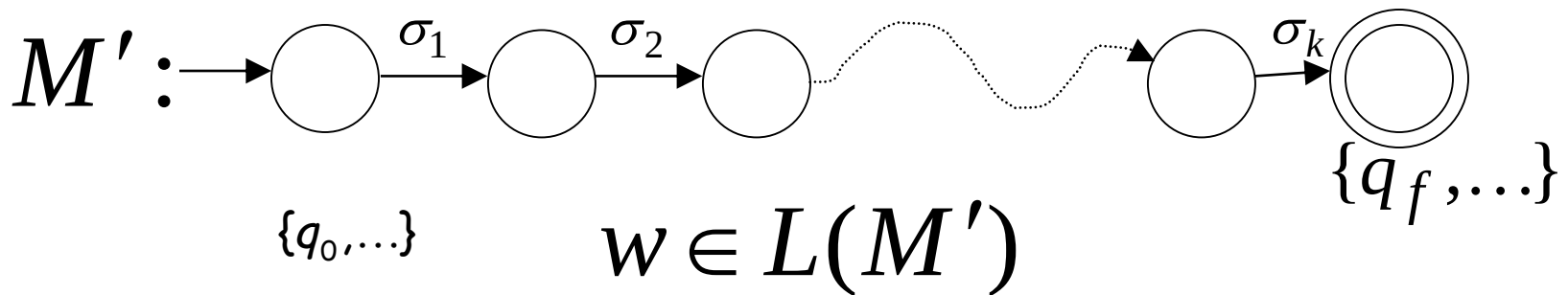
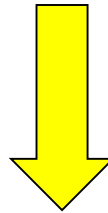
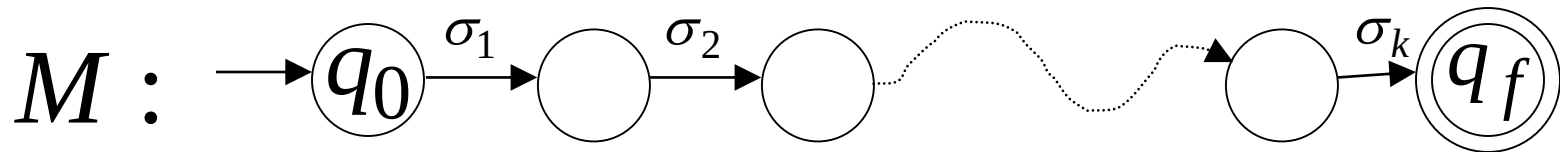
Induction Step: $|v| = k + 1$

$$v = \underbrace{a_1 a_2 \cdots a_k}_{v'} a_{k+1} = v' a_{k+1}$$



Therefore if $w \in L(M)$

$$w = \sigma_1 \sigma_2 \cdots \sigma_k$$

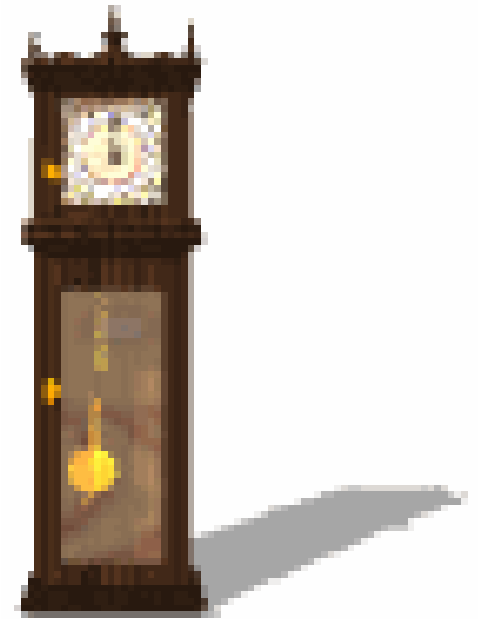


We have shown: $L(M) \subseteq L(M')$

We also need to show: $L(M) \supseteq L(M')$

(proof is similar)

Activity Time



NFAs vs. DFAs

NFAs vs. DFAs

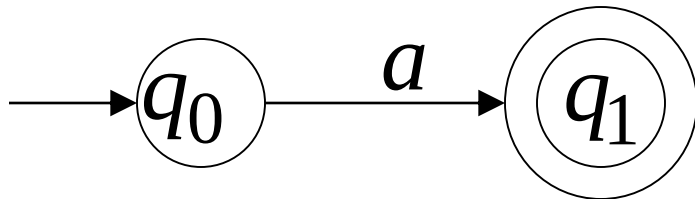
What is non-determinism?

Which is useful?

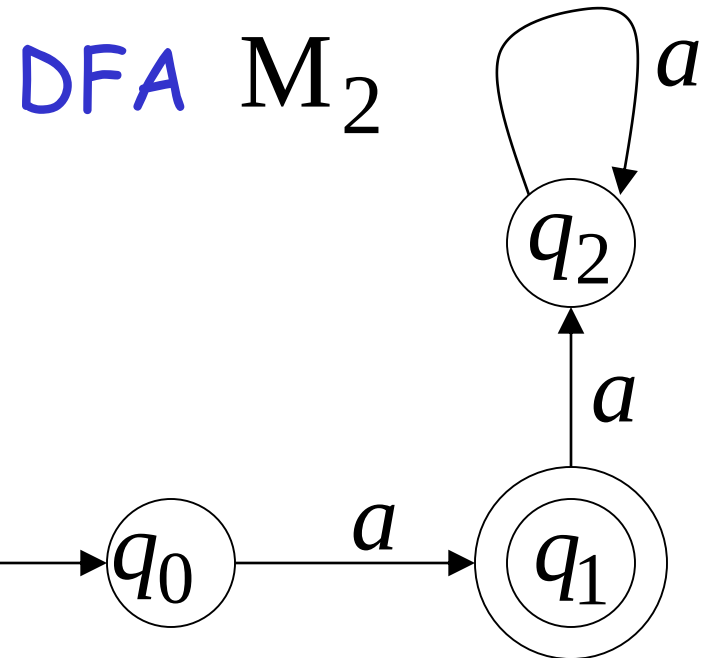
Which is powerful?

- NFAs are interesting because we can express languages easier than DFAs

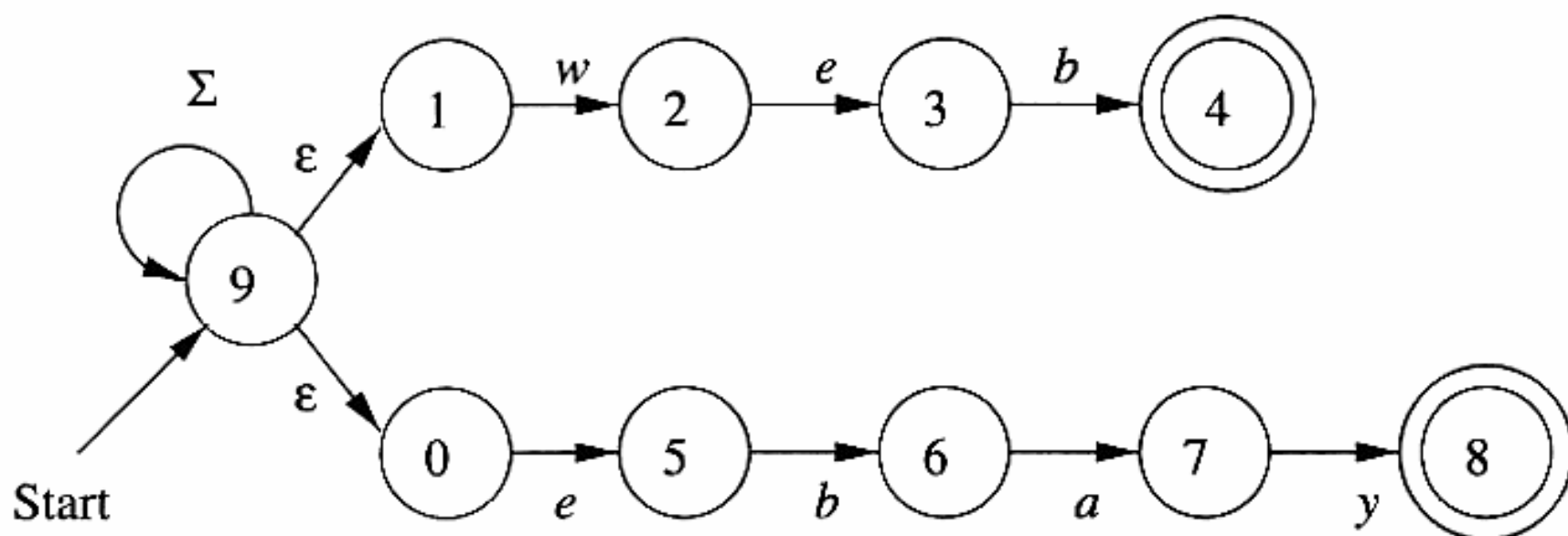
NFA M_1



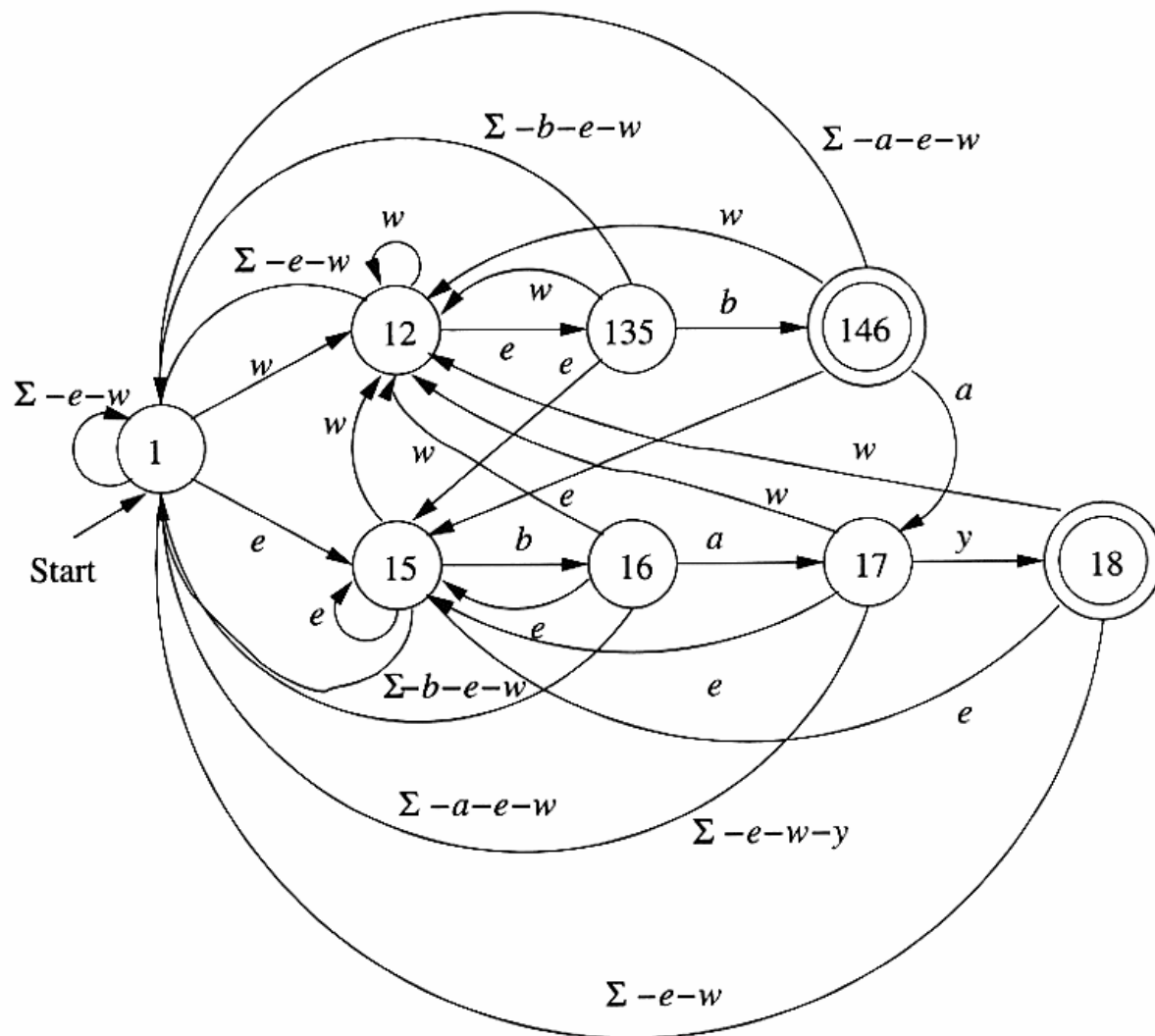
$$L(M_1) = \{a\}$$



$$L(M_2) = \{a\}$$



An NFA that searches for the words **web** and **ebay**



Conversion of the NFA to a DFA